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MILITARY SPECIFICATION

ROCKET MOTOR M119, ASSEMBLY

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Rocket Motor M119 Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, (see 6.2.1) the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-P-82676	Propellant, N-53

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-286	Propellants, Solid; Sampling, Examination and Testing
MIL-STD-414	Sampling Procedures and Tables for Inspection By Variables for Percent Defective
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC NO. N 4385 FSC 1377

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MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1949	Inspection, Magnetic Particle
DOD-STD-2101	Classification of Characteristics
MS17983	Compass-Magnetic, Pilots Standby

2.1.2 Drawings. The following government drawings, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Ordnance Station, Indian Head, Maryland (Code Ident 14083)

DL11740301	Rocket Motor M119 Assembly

Naval Sea Systems Command (Code Ident 53711)

5184336	Box, Packing, Fiberboard (Inner) M119 Rocket Motor
5184337	Box, Packing, M119 Rocket Motor
5184338	Box, Fiberboard (Outer) Alternate, Packing M119 Rocket Motor

(Copies of specifications, standards, and drawings, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article sample. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.1.1 Propellant aging. Propellant samples shall be prepared and tested in accordance with the requirements of 4.4.1. Tests shall be performed in accordance with MIL-P-82676. Failure of the propellant to meet the requirements of MIL-P-82676 shall be cause for rejection of the batch.

3.1.2 Propellant morphology. Propellant grain assemblies shall be examined in accordance with the requirements of 4.4.2. Test results will be used to determine propellant homogeneity, particle size distribution, and presence of oxidizer agglomerates, binder rich areas, uncured binder, or other anomalies in propellant morphology. Existence of such conditions, which may result in unacceptable propellant performance or aging characteristics, shall be cause for rejection of the batch.

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3.1.3 Environmental requirements. The rocket motor assemblies shall withstand the environmental conditioning requirements of 3.1.3.1 through 3.1.3.5. Rocket motor assemblies subjected to this conditioning shall show no evidence of disintegration of propellant or derangement of components under visual and radiographic examinations and shall meet the ballistic requirements of 3.3.8 when test fired in accordance with 4.4.4.

3.1.3.1 High temperature. The rocket motor shall withstand the high temperature test of 4.4.3.1.

3.1.3.2 Low temperature. The rocket motor shall withstand the low temperature test of 4.4.3.2.

3.1.3.3 Temperature shock. The rocket motor shall withstand the temperature shock test of 4.4.3.3.

3.1.3.4 Temperature-altitude-humidity. The rocket motor shall withstand the temperature-altitude-humidity test of 4.4.3.4.

3.1.3.5 Vibration. The rocket motor shall withstand the vibration test of 4.4.3.5.

3.1.4 Static firing. Within 30 days after completion of environmental testing, the rocket motors shall be static fired for comparison with the requirements of 3.3.8 and table II.

3.2 Materials and parts. All materials, parts, and assemblies shall be in accordance with the documents listed on DL11740301.

3.2.1 Propellant. The propellant shall conform to MIL-P-82676.

3.2.2 Firing pin release. The firing pin of the firing pin assembly, P/N 11740310, less safety pin, shall release from the cocked position when the application of an axial load of 8.5 to 20.5 pounds is applied to the sear.

3.3 Assemblies. Assemblies shall conform to the requirements listed on DL1174031. The contractor shall retain for the service life of the unit propellant production records including batch number, mix designation, rocket motor lot number, rocket motor serial number, igniter lot number, and igniter serial number. These shall be cross referenced to the assembled rocket motor lot number and to specific rocket motor serial numbers (see 6.2.1).

3.3.1 Age limitations. The following age limitations shall be observed.

3.3.1.1 Explosive components. All explosive components, including but not limited to ignition powders and igniters, shall be manufactured or recertified within 12 months of the original contract scheduled delivery date.

3.3.1.2 Primers. Primers shall be manufactured and certified within 24 months or recertified within 12 months of the original contract scheduled delivery date.

3.3.1.3 Propellant. The propellant shall be manufactured not more than 150 days prior to the orginal contract scheduled delivery date for the rocket motor lot.

3.3.2 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.3.3 Identification marking. Each rocket motor shall be identified as specified on Drawing 11740301.

3.3.3.1 Production lot designation. The production lot number shall be in accordance with MIL-STD-1168. The construction of the lot number shall be as follows:

ABC-85-D-018-124
a b c d e

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where

a = Manufacturer's identification code
b = The year of manufacture of the oldest batch of propellant in the lot
c = The month of manufacture of the oldest batch of propellant in the lot as an alpha code in accordance with MIL-STD-1168
d = Lot interfix number (controlled by Naval Ordnance Station, Indian Head, MD 20640-5000, or the procuring activity.)
e = Lot sequence number—lot number (manufacturer is responsible for the assignment of lot numbers)

3.3.4 Torque. Torque requirements shall be as specified in table I. No loosening or deformation shall occur upon application of the breakaway torque.

TABLE I. Breakaway torque.

Components	Seating or bottoming torque (inch-pounds)	Breakaway torque (inch-pounds, minimum)
Cap, End, 11740330 to Propellant Grain Assembly, 11740307	500 ± 100	200
Cap, Head End, 11740304 to Propellant Grain Assembly, 11740307	500 ± 100	200
Pin, Firing Assembly, 11740310 to Cap, Head End, 11740304	300 ± 50	200

3.3.5 Residual magnetism. The rocket motor assembly shall not deflect the indicator of the compass more than 5 degrees in either direction when tested as specified in 4.5.10.

3.3.6 Leak test. The sealed rocket motor assembly shall not exhibit a leak rate of greater than 1×10^{-3} cubic centimeters (cm^3) of helium per second (see 4.5.11). Motors exceeding this leakage rate shall be rejected. Rejected motors may be reworked and retested to this requirement.

3.3.7 Radiographic. Each rocket motor assembly shall be radiographically inspected (see 4.5.12). Internal configuration and condition shall be as specified on applicable drawings. The radiographs shall be retained for 6 years or one year past the service life of the unit, whichever is longer (see 6.2.1).

3.3.8 Ballistic. The rocket motor, loaded with acceptable explosive components, shall function without mechanical failure (see 6.6.1) and shall perform in accordance with the requirements of table II. The rocket motors shall meet the following requirements when test fired (see 4.4.4 and 4.5.13):

a. No unit component shall be ejected
b. No component subjected to loading shall fail
c. No excessive nozzle erosion, hot spot or burn-through shall occur.
d. No failure to fire shall occur.

TABLE II. Rocket motor performance.

Parameter[1]	−65° to −70°F		65° to 70°F		160° to 165°F	
	Min	Max	Min	Max	Min	Max
Peak thrust (lbs)	1200	1800	1500	2300	1700	2700
Total impulse (lb-sec)	370	430	370	430	370	430
Action time (sec)	0.280	0.420	0.230	0.350	0.180	0.320
Burning time (sec)	0.220	0.310	0.180	0.270	0.140	0.250
Ignition delay time (sec)	0.000	0.015	0.000	0.015	0.000	0.015

[1]See 6.6 for definitions.

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3.3.8.1 Post static firing disassembly. A post static firing disassembly and inspection shall be conducted on all first article and lot acceptance test units.

3.4 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following.

3.4.1 Metal defects. The component shall be free from cracks, splits, cold shots, inclusions, porosity or similiar defects.

3.4.2 Burrs. No part shall have a burr which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.4.3 Foreign matter. The component or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter. There shall be no excess sealant on surfaces adjacent to the sealed parts.

3.4.4 Magnetic defects. Each part shall pass magnetic particle inspection in accordance with applicable drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 65° F to 95° F
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum

4.4 First article inspection. Unless otherwise specified in the contract or order (see 6.2.1), a first article sample of 21 loaded rocket motor assemblies, shall be submitted to the Naval Ordnance Station, Indian Head, MD 20640-5000 for first article inspection. The first article samples shall be subjected to the quality conformance inspection of 4.5, with the exception of the test firing of 4.5.13, followed by environmental testing, radiographic examinations and static firing in the sequence specified in table III. Acceptance of the first article sample shall be based on no defects in the sample (see 6.3).

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TABLE III. First article loaded assembly inspection.[1]

Test	Unit number																				
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Radiographic inspection (4.5.12)	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a
Vibration, 165°F (4.4.3.5)	b									b		b									
Vibration, −65°F (4.4.3.5)		b	b								c										
Temperature shock (4.4.3.3)				b	b	b				c	b	c									
Temperature-altitude-humidity (4.4.3.4)							b				d	d									
High temperature (4.4.3.1)								b		e	e	f									
Low temperature (4.4.3.2)									b	d	f	e									
Radiographic inspection (4.5.12)	c	c	c	c	c	c	c	c	c	f	g	g									
Static fire, 165°F (4.4.4)	d				d	d		d		g		h				b	b	b			
Static fire, 70°F (4.4.4)													b	b	b						
Static fire, −65°F (4.4.4)		d	d	d			d		d		h								b	b	b
Disassembly (4.5.13.1)	e	e	e	e	e	e	e	e	e	h	i	i	c	c	c	c	c	c	c	c	c

[1]Letters denote sequence of testing.

4.4.1 Propellant aging. Two 1-gallon blocks of cured propellant from each batch of propellant used in the first article sample shall be submitted to the Naval Ordnance Station, Indian Head, Maryland, 20640-5000, for age testing. High temperature aging shall be conducted for 84 days at 160°F with an atmosphere of 22 percent relative humidity and with dry air (a lower temperature will be used if there is supporting evidence that the degradation is at a faster rate at a lower temperature). At the end of the 84 day aging, physical properties shall conform to MIL-P-82676. Tests shall also be performed at 0, 28, and 56 days to determine trends. The pull rate of stress/strain determination shall be 2.0 inches/minute. The sample shall be cast into cylindrical cardboard cartons approximately 7 inches in diameter. All samples shall be bare propellant individually wrapped in moisture proof packaging and properly identified as to propellant lot and batch number.

4.4.2 Propellant morphology. Four propellant grain assemblies from each batch used in the first article sample shall be submitted to the Naval Ordnance Station, Indian Head, Maryland, 20640-5000, for morphological examination of the propellant. This examination shall include, but is not limited to, microscopy and chromatography. All samples shall be individually wrapped in moisture proof packaging and properly identified as to propellant lot, batch number, and grain serial number.

4.4.3 Environmental tests.

4.4.3.1 High temperature. The rocket motor shall be subjected to high temperature storage in accordance with Method 501, Procedure I, of MIL-STD-810. The test item shall be maintained at 165° F for a minimum of 50 hours.

4.4.3.2 Low temperature. The rocket motor shall be subjected to low temperature storage in accordance with Method 502, Procedure I, of MIL-STD-810. The test item shall be maintained at −65° F for a minimum of 50 hours.

4.4.3.3 Temperature shock. The rocket motor shall be subjected to a temperature shock test in accordance with the Method 503, of MIL-STD-810. The duration of each cycle shall be 8 hours. The low temperature and high temperature are −65° F and 165° F, respectively.

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4.4.3.4 Temperature-altitude-humidity. The rocket motor shall be subjected to a temperature-altitude-humidity test in accordance with Method 518 of MIL-STD-810B and as modified herein (see 6.2.1):

Step 3—Use 40,000 feet in lieu of 50,000 feet.
Step 10—Repeat steps 2 through 9 ten times.
Step 13—Use 40,000 feet in lieu of 50,000 feet.

4.4.3.5 Vibration. The rocket motor assembly shall be subjected to a vibration test in accordance with Method 514, Category (b) of MIL-STD-810B as specified in table 514-I Procedure I, Parts 1, 2 and 3 and Curve D (see 6.2.1). Before vibration, the units shall be temperature conditioned in accordance with table III for a minimum of 8 hours. If the rocket motor must be removed from the specified temperature limits for a period of more than 5 minutes, the testing shall be stopped and the unit shall be reconditioned. A reconditioning time of 2 minutes for every minute the unit is outside the specified temperature envelope shall be required. A continuous temperature history of the unit shall be kept for all conditioning and testing time. Unless otherwise noted, the rocket motor shall be mounted on a test fixture in a manner that will simulate actual seat/aircraft installation and excited through the attachment fittings.

4.4.4 Static firing test for first article. Within 30 days after completion of environmental testing, the rocket motor shall be static fired at the Naval Ordnance Station, Indian Head, MD 20640-5000. The motors shall be temperature conditioned per table III for a minimum of 8 hours and test fired within 10 minutes after removal from the conditioning chamber. The rocket motor assembly shall be mounted in the test fixture in a horizontal position. Initiation of the assembly shall simulate aircraft initiation by the use of a lanyard attachment to the units firing mechanism and a pull type solenoid. The solenoid shall have a lift force capability of 50 pounds at the end of a 2-inch stroke. Data specified in table II shall be analyzed in accordance with MIL-STD-414, standard deviation method single or double specification limit (as applicable) variability unknown with an AQL of 2.5 percent. Minimum equipment accuracies shall be as shown in table IV. Any failure of a rocket motor assembly to comply with the requirements of 3.3.8 and table II shall be cause for rejection of the first article sample.

TABLE IV. Test equipment accuracies.

Requirement	Equipment accuracy
Maximum thrust	±2.0 percent
Weight	±0.005 pounds
Time	±0.001 second
Maximum acceleration	±5.0 percent
Frequency	±1.0 percent of measured value or ±1 hertz, whichever is greater
Temperature	±2°F
Recording instruments	Permanent records

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the examinations and tests of 4.5.3 through 4.5.13 conducted on samples selected as specified in 4.5.1.3. The contractor is responsible for assuring that all rocket motors offered for acceptance meet all the requirements of the applicable drawings and specifications. Items of inspection shall be serialized and inspection data traceable to a specific serial number.

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4.5.1 Sampling for quality conformance inspection.

4.5.1.1 Inspection lot. An inspection lot is defined in 6.6.8.

4.5.1.2 Sample for examination. Designation of defects (critical, major, and minor) shall be in accordance with DOD-STD-2101.

a. Critical and major defects: Each rocket motor assembly shall be examined for all critical and major defects.
b. Minor defects: Samples for examination shall be selected in accordance with inspection level II of MIL-STD-105.

4.5.1.3 Sampling for firing test. A sample of loaded rocket motor assemblies shall be submitted to the Naval Ordnance Station. Indian Head, MD 20640-5000, for static firing in accordance with 4.5.13. The sample size for individual lots shall be in accordance with MIL-STD-414, inspection level III, for each temperature tested except that each batch of propellant must be represented by a minimum of one unit per temperature extreme. Samples shall be chosen by Naval Ordnance Station, Indian Head, Maryland personnel or by a designated Government representative upon completion of the production lot assembly. Loaded rocket motor samples shall not be preselected during production. Rocket motors selected for samples shall not be included as a part of the quantity specified for delivery in the contract or order. The total lot acceptance sample shall be the total ballistic sample plus one unit for detailed breakdown. Table V outlines that ballistic sample size.

TABLE V. Sample size for firing test.

Lot size	Lot acceptance sample (minimum)	Total ballistic sample (minimum)	Sample per temperature extreme
Up to 40	7	6	3
41—65	9	8	4
66—110	11	10	5
111—180	15	14	7
181—300	21	20	10
300—340	23	22	11
341—360	25	24	12

4.5.2 Resubmission of rejected lot. A sample of rocket motor assemblies from a rejected lot shall be selected in accordance with 4.5.1.3 and submitted when, with prior approval of the acquiring activity, the contractor requests that the rejected lot be test fired under different conditions or the contractor believes a rejected lot may be reworked to meet the requirements of this specification.

4.5.3 Examinations.

4.5.3.1 Critical and major defects. Each rocket motor assembly shall be visually and radiographically examined for critical and major defects specified in 4.5.1.2 and the applicable drawings. Any rocket motor assembly found to contain a critical or major defect shall be rejected (see 6.2.2).

4.5.3.2 Minor defects. Sample rocket motor assemblies selected in accordance with 4.5.1.2.b shall be examined for minor defects in accordance with the applicable drawings, 3.3.2, 3.3.3, and 3.4. The acceptable quality level (AQL) shall be 2.5 defects per hundred units (see 6.2.2).

4.5.4 Propellant tests. The following modifications shall be made to the testing requirements of MIL-P-82676. For each propellant production lot, the manufacturer, lot number, and date of manufacture for the materials and components of each propellant batch shall be identified. The burning rates at 500, 1000, and 1500 pounds per square inch shall be determined at −65°F, 70°F, and 165°F for each pressure (three samples at each temperature and pressure per batch). The manufacturer shall perform the strand burning rate tests at the temperatures and pressures stated above in accordance with MIL-STD-286, Method T803.1, on each batch of propellant. The propellant strands shall be 6 inches long and 1/4 to 3/8 inch in diameter. Tensile strengths and elongations at −65°F, 70°F, and 165°F shall be determined at a pull rate of 2 inches/minute for three samples at each temperature per batch. The heat of explosion shall be determined for three samples per batch. Mix sheets shall be used to verify that ingredients were properly weighed and added (see 6.2.2).

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4.5.5 Firing pin release. Each firing pin assembly shall be subjected to a firing pin release test prior to being assembled to the rocket motor. The firing pin assembly shall be secured in an approved fixture. A pull shall be applied to the sear in a longitudinal direction in conjunction with the fixture which registers the amount of pull. Failure of any assembly to comply with the requirements of 3.2.2 shall be cause for rejection of the assembly (see 6.2.2).

4.5.6 Age limitations. Age limitations of all assemblies shall be verified by the examination of production records for conformance to 3.3.1.

4.5.6.1 Explosive components. The ages of explosive components such as ignition primers and igniters for each assembly shall be verified by examination of production records for conformance to 3.3.1.1 (see 6.2.2).

4.5.6.2 Primers. The ages of primers for each assembly shall be verified by the examination of production records for conformance to 3.3.1.2 (see 6.2.2).

4.5.6.3 Propellant. The age of the propellant in each assembly shall be verified by the examination of production records for conformance to 3.3.1.3 (see 6.2.2).

4.5.7 Protective finish. All assemblies shall be visually examined for conformance to 3.3.2 (see 6.2.2).

4.5.8 Identification marking. All assemblies shall be visually examined for conformance to 3.3.3 (see 6.2.2).

4.5.8.1 Production lot designation. All assemblies shall be visually examined for conformance to 3.3.3.1 (see 6.2.2).

4.5.9 Breakaway torque. A breakaway torque shall be applied to each rocket motor threaded joint specified in table I and shall be used to determine the adequacy of the joints. The assembly shall be secured in a fixture prior to the application of the torque. Any failure of a rocket motor to comply with the requirements of 3.3.4 shall be cause for rejection of the rocket motor (see 6.2.2)

4.5.10 Residual magnetism. The test shall be performed in an area free from local magnetic effects. Each rocket motor assembly shall be slowly passed 6 inches in front of an MS17983 compass along the entire length on the rocket motor. If the compass indicator deflects more than 5° in either direction, the rocket motor assembly shall be rejected (see 6.2.2).

4.5.11 Leak test. Each rocket motor shall be subjected to a 100 percent leak test. The rocket motor shall be subjected to a helium diffusion leak test as follows:

a. Place rocket motors in a vacuum chamber and decrease chamber pressure to 1 inch Hg maximum for 25 minutes.

b. Break chamber vacuum pressure by back filling the chamber with helium and pressurizing to 15 psig for 5 minutes minimum.

c. Reduce chamber pressure to ambient conditions and place the rocket motors immediately into a helium atmosphere for 5 minutes.

d. Remove the rocket motors from the helium atmosphere and remove superficially entrapped helium from the rocket motors using dry nitrogen at 20 psig.

e. Place the rocket motors in the leak detector vacuum chamber.

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f. All rocket motors must be leak tested within 20 minutes after removal from the pressurized chamber.

g. All rocket motors whose leakage rate is in excess of 1×10^{-3} standard cm^3/sec air shall be rejected.

Any failure of a rocket motor to comply with the requirements of 3.3.6 shall be cause for rejection of the rocket motor (see 6.2.2).

4.5.12 Radiographic inspection. Each rocket motor shall be radiographically inspected in accordance with MIL-STD-453, radiographic quality level 1. The propellant grain assembly shall be X-rayed twice along the longitudinal axis with views 90° apart. Any deficiencies shall be cause for rejection of the unit. Radiographs shall be capable of detecting defects and rocket motor components 0.03 inch in length or width, which ever is smaller. If all components cannot be shown using standard radiographic techniques, neutron radiography shall be employed to show these components. All radiographs shall be inspected on-site by a designated Government representative. Acceptance of the loaded rocket motor lot shall be subject to approval of the radiographs by the designated Government activity.

4.5.13 Production lot acceptance firing. Rocket motors selected in accordance with 4.5.1.3 shall be static fired at the Naval Ordnance Station, Indian Head, Maryland 20640-5000 within 30 days after submission for acceptance of the rocket motor lot (see 6.2.1). The motors shall be divided into two groups (and conditioned for 8 hours minimum) as follows: Group 1, −65°F to −70°F; Group 2, 160°F to 165°F. Static firing shall be within 10 minutes after removal from the conditioning chamber. The rocket motor assembly shall be mounted in the test fixutre in a horizontal position. Initiation of the assembly shall simulate aircraft initiation by the use of lanyard attached to the units firing mechanism and a pull type solenoid. The solenoid shall have a lift force capability of 50 pounds at the end of a 2-inch stroke. Data specified in table II shall be analyzed in accordance with MIL-STD-414, standard deviation method, single or double specification limit (as applicable), variability unknown with an AQL of 2.5 percent. Minimum equipment accuracies shall be as shown in table IV. Any failure of a rocket motor assembly to comply with the requirements of 3.3.8 and table II shall be cause for rejection of the lot represented by the sample.

4.5.13.1 Post static firing disassembly. A post firing disassembly and inspection shall be conducted on all first article and lot acceptance test units. Detailed observations on conditions including marginal conditions of individual components shall be noted. All firings shall be evaluated against the requirements of 3.3.8.

4.5.14 Workmanship. All parts and assemblies shall be inspected for conformance to applicable documents and drawings.

4.5.14.1 Metal defects. All components shall be visually examined for conformance to 3.4.1 (see 6.2.2).

4.5.14.2 Burrs. All parts shall be visually examined for conformance to 3.4.2 (see 6.2.2).

4.5.14.3 Foreign matter. All components and assemblies shall be examined by an appropriate method for conformance to 3.4.3 (see 6.2.2).

4.5.14.4 Magnetic particle inspection. When indicated on the drawing, each part shall be 100 percent magnetic particle inspected in accordance with MIL-STD-1949. Any deviation from 3.4.4 shall result in rejection of the defective part (see 6.2.2).

4.6 Packaging inspection. The packaging, packing, and marking shall be inspected to verify conformance with the requirements of Section 5.

5. PACKAGING

5.1 Packing and marking.

5.1.1 Level A. Packing and marking shall be in accordance with Drawings 5184336 and 5184337.

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5.1.2 Level B. Packing and marking shall be in accordance with Drawings 5184336 and 5184338.

5.2 Ammunition data cards. As specified in the contract, ammunition data cards (DD Form 1650) completed in accordance with MIL-STD-1167 shall be included with each lot of loaded rocket motors. One copy shall be forwarded to the Commanding Officer, Naval Ordnance Station, Indian Head, Maryland 20640-5000 (see 6.2.2).

6. NOTES

6.1 Intended use. The M119 Rocket Motor is intended for use as a stabilization motor to maintain the Mk-J5D ejection seat in the required attitude to ensure proper drogue chute deployment.

6.2 Ordering data.

6.2.1 Acqusition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specification
b. Quantity required (exclusive of first article and static-firing test samples)
c. That MIL-STD-810B shall be used for the temperature-altitude-humidity and the vibration tests (see 2.1.1, 4.4.3.4, and 4.4.3.5)
d. Whether a first article sample is required (see 3.1, 4.4, and 6.3)
e. That the contractor shall retain production records (see 3.3)
f. That the contractor retain radiographs on file for 6 years or 1 year past the service life of the unit, whichever is longer (see 3.3.7)
g. Whether the contractor is responsible for testing and whether he shall use his own facilities (see 4.1, 4.4.4, and 4.5.13)
h. Inspection conditions, if other than as specified (see 4.3)
i. Disposition of hardware and scrap propellant (see 6.5)
j. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DoD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP5 are applicable.

6.2.2 Data requirements. When this document is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.5.4, 4.5.5, 4.5.10, 4.5.11, 4.5.14.4	Test Report	DI-T-2072	Contractor format
4.5.3.1, 4.5.3.2, 4.5.6.1, 4.5.6.2, 4.5.6.3, 4.5.7, 4.5.8, 4.5.8.1, 4.5.9, 4.5.14.1, 4.5.14.2, 4.5.14.3	Certification Data/Report	UDI-A-23264B	
5.2	Ammunition Data Cards	DI-MISC-80043	

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(Data Item Descriptions related to this specification, and identified in this section will be approved and listed as such in DoD 5010.12L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the M119 rocket motor assembly should be a first article sample. The first article should consist of 21 units, the contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples of first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Hazard information.

6.4.1 Loaded assembly hazard information.

Explosive Hazard Classification: Class 1, Div. 3
Storage Compatibility: Group C
Department of Transportation (DOT) Hazard Class: Class B
DOT Marking: Explosive Power Device Class B
Handle Carefully—(Keep Fire Away)

6.4.2 Hazard notice. The rocket motor assembly described herein and certain of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these hazards and take appropriate measures to guard against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.5 Disposition of hardware, scrap propellant and metal parts. The contract shall specify that all items purchased become the property of the U.S. Government; no hardware, propellant, or metal parts (acceptable, scrapped or tested) shall be disposed of without authorization of the procuring activity: All items shall be made available to the procuring activity upon request. Disposal of any tested hardware, propellant, or metal part without authorization of the procuring activity will result in the test being declared invalid and a retest being required.

6.6 Definitions.

6.6.1 Mechanical failure. Mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.6.2 Ignition delay. Ignition delay is defined as the time that elapses between the firing pin release to the point at which the rocket thrust reaches 10 percent of its maximum value.

6.6.3 Burn time. Rocket motor burn time is defined as the time from 10 percent thrust on the increasing portion of the thrust-time curve until the thrust-time curve breaks sharply downward at web burnout, using the tangential method.

6.6.4 Action time. Action time is defined as the time between the points where the rocket thrust reaches 10 percent of its maximum value at the start of burning and falls below 10 percent of the maximum value at the end of burning.

6.6.5 Total impulse. Total impulse is defined as the area under the rocket thrust-time curve bounded by the action time limits.

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6.6.6 Propellant batch. A propellant batch is a mix of ingredients, each ingredient composed entirely of one homogeneous lot, that is mixed at one time in one vessel, resulting in a homogeneous mixture.

6.6.7 Propellant lot. A propellant lot is a group of one or more propellant batches, all using the same lot of each ingredient.

6.6.8 Inspection Lot. An inspection lot shall consist of an identifiable group of assembled units containing: metal parts assemblies produced by a homogeneous process, a single propellant lot consisting of one or more propellant batches, and a single lot of accepted igniters.

6.7 Subject term (key word) listing:

Attitude control
Drogue chute
Ejection seat
Rocket motor
Testing

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - OS

Project 1377-N906

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-50753A(OS)	ROCKET MOTOR M119, ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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